Design Milestones-Egg Gurdian

Problem Understanding:

The goal of the Egg Drop Challenge was to design a structure that protects a raw egg from breaking when dropped from a height. The design had to be made only with straws and string without using glue or tape.

Idea Exploration:

Different structural ideas were explored such as cages, suspension systems, and tripod structures. The main focus was creating a design that could distribute impact forces and stabilize the egg during the fall.

Development

A triangular support structure was selected because triangles are structurally stable and distribute forces effectively. Multiple legs were added to absorb and spread the impact energy.

Prototype Creation

Two prototypes were developed. The first one used green straws tied with strings to create a lightweight structure. The second prototype explored a more compact triangular protective frame.

Final Design Decision

The final design, named **Egg Guardian**, protects the egg by placing it in the center of a triangular structure supported by extended legs that absorb the shock during landing.

Keywords While Building Prompts

- Impact absorption
- Structural stability
- Triangular geometry
- Lightweight structure
- Energy distribution
- Protective frame
- Egg protection

Selected Impactful Prompts

Example prompts used during the design thinking process:

“How can I protect a egg using only straws and string?”

“What structural shapes are the most stable for absorbing impact?”

“How can a tripod structure distribute forces during a fall?”

“What simple geometry can increase structural strength?”

AI Role Tags

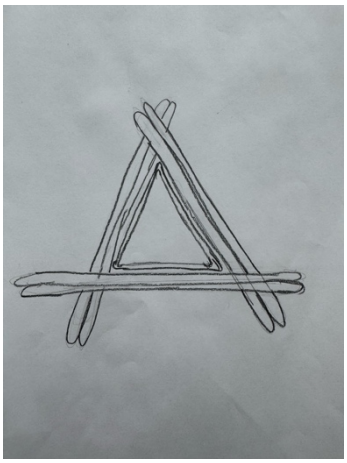
Clarifier
AI helped clarify the design requirements and constraints of the assignment.

Re-framer
AI suggested thinking about the problem as an impact absorption system rather than only a container for the egg.

Expander
AI expanded design possibilities by suggesting tripod structures and triangular frames.

Critique
AI evaluated the stability of the structure and suggested improving the distribution of forces.

Design Decisions



- The egg was placed **in the center of the structure** for maximum protection.
- A **triangular frame** was used for structural stability.
- **Extended legs** were added to absorb and distribute the impact force.
- **String connections** were used to strengthen joints without glue or tape.

Accepted, Rejected, and Transformed Ideas

Accepted

- Triangular structure
- Multi-leg support system
- Central egg placement

Rejected

- Fully closed cage designs
- Flat base structures with low shock absorption

Transformed

- Simple straw structures were transformed into a **tripod-based protective frame** with multiple impact-absorbing legs.

Reflection

This project showed how simple materials can be used to create effective protective structures through geometric thinking. By focusing on balance, force distribution, and structural stability, the Egg Guardian design was able to protect the egg while remaining lightweight. The design process also highlighted the importance of experimentation and iterative thinking when solving physical design problems.

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